

HRW 12e Conceptual Practice Quiz — Instructor Version

Instructions

Multiple choice, one correct answer per question. Unless otherwise stated, symbols have their usual meaning.

Reference links are provided for deeper review.

EM Induction (Q1–Q10)

Q1. What does Lenz's law state about the direction of an induced current?

- (a) It enhances the change in flux
- (b) It is independent of flux changes
- (c) It always flows clockwise
- (d) It opposes the change in flux that produced it

Answer: (d)

Explanation: Lenz's law says the induced current creates a magnetic effect that opposes the change in magnetic flux, consistent with energy conservation.

Resources: OpenStax Univ. Phys. Vol. 2 – 13.1 Faraday's Law: <https://openstax.org/books/university-physics-volume-2/pages/13-1-faradays-law>

Q2. According to Faraday's law, the magnitude of the induced emf in a loop is proportional to:

- (a) The magnetic field strength only
- (b) The loop resistance
- (c) The rate of change of magnetic flux
- (d) The loop circumference

Answer: (c)

Explanation: Faraday's law gives $|\mathcal{E}| = |d\Phi_B/dt|$, so the emf depends on how fast the flux changes.

Resources: Same as Q1

Q3. Which expression correctly defines magnetic flux through a flat loop area A in field B at angle θ ?

- (a) $\Phi_B = BA$
- (b) $\Phi_B = B/A$
- (c) $\Phi_B = BA \cos \theta$
- (d) $\Phi_B = BA \sin \theta$

Answer: (c)

Explanation: Magnetic flux is $\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \theta$.

Resources: Same as Q1

Q4. In motional emf, charges in a conductor moving with velocity $\vec{v}$ through field $\vec{B}$ are separated by which force?

- (a) $q\vec{E}$
- (b) Gravity
- (c) $q\vec{v} \times \vec{B}$
- (d) Nuclear force

Answer: (c)

Explanation: The magnetic part of the Lorentz force drives charge separation and produces the emf.

Resources: Same as Q1

Q5. A loop enters a region of increasing magnetic field into the page. To oppose the change, the induced current's own magnetic field points:

- (a) Into the page
- (b) Out of the page
- (c) Horizontally
- (d) It is zero

Answer: (b)

Explanation: The increasing into-page flux is opposed by an induced field out of the page (Lenz's law).

Image: images/em_induction_loop.png

Resources: Same as Q1

Q6. The negative sign in Faraday's law $\mathcal{E} = -d\Phi_B/dt$ expresses:

- (a) A meaningless sign convention
- (b) That emf is always negative
- (c) Lenz's law: induced emf opposes flux changes
- (d) That flux is always decreasing

Answer: (c)

Explanation: The minus sign encodes that induced currents oppose the change that produces them.

Resources: Same as Q1

Q7. Eddy currents are:

- (a) DC currents in batteries
- (b) Loops of induced current in bulk conductors
- (c) Only currents in superconductors
- (d) Hypothetical with no real effects

Answer: (b)

Explanation: Time-varying flux induces circulating currents that cause heating and magnetic braking.

Resources: Same as Q1

Q8. Which situation will NOT induce an emf in a conducting loop?

- (a) Changing the magnetic field through the loop
- (b) Changing the loop's area in a magnetic field
- (c) Rotating the loop in a magnetic field
- (d) Keeping field, area, and orientation all constant

Answer: (d)

Explanation: No change in magnetic flux means no induced emf.

Resources: Same as Q1

Q9. A transformer operates primarily due to:

- (a) Coulomb's law
- (b) Faraday's law of induction
- (c) Gauss's law for magnetism
- (d) Ohm's law

Answer: (b)

Explanation: A changing flux in the core induces emfs in both primary and secondary windings.

Resources: Same as Q1

Q10. The SI unit of magnetic flux is:

- (a) Tesla
- (b) Weber
- (c) Henry
- (d) Volt

Answer: (b)

Explanation: One weber equals one tesla–square-meter.

Resources: Same as Q1

LR Circuits & Inductors (Q11–Q20)

Q11. In an RL circuit, the time constant τ is:

- (a) R/L
- (b) L/R
- (c) RC
- (d) $1/RC$

Answer: (b)

Explanation: For an RL circuit $\tau = L/R$, which sets how fast current approaches steady state.

Resources: OpenStax Vol. 2 – 14.4 RL Circuits: <https://openstax.org/books/university-physics-volume-2/pages/14-4-rl-circuits>

Q12. Immediately after an RL circuit is connected to a DC source, the ideal inductor:

- (a) Allows full current instantly
- (b) Opposes any change in current, acting like an open circuit
- (c) Behaves like a capacitor
- (d) Has no effect

Answer: (b)

Explanation: An inductor resists sudden change in current, so initial current is limited.

Resources: Same as Q11

Q13. Long after connection to a DC source, the ideal inductor in an RL circuit behaves like:

- (a) An open circuit
- (b) A short circuit (wire)
- (c) A large resistor
- (d) A capacitor

Answer: (b)

Explanation: Once current is steady, $dI/dt = 0$ so the inductor's voltage drop is zero.

Resources: Same as Q11

Q14. The energy stored in an inductor is:

- (a) LI
- (b) $\frac{1}{2}LI^2$
- (c) $\frac{1}{2}CV^2$
- (d) RI^2

Answer: (b)

Explanation: Magnetic energy in an inductor is $U = \frac{1}{2}LI^2$.

Resources: Same as Q11

Q15. The SI unit of inductance is:

- (a) Tesla
- (b) Weber
- (c) Henry
- (d) Ohm

Answer: (c)

Explanation: One henry is one weber per ampere.

Resources: Same as Q11

Q16. The induced emf across an inductor is:

- (a) $\mathcal{E} = LI$
- (b) $\mathcal{E} = -L dI/dt$
- (c) $\mathcal{E} = IR$
- (d) $\mathcal{E} = qvB$

Answer: (b)

Explanation: This shows that the inductor opposes changes in current.

Resources: Same as Q11

Q17. Increasing L (with R fixed) in an RL circuit makes the current reach steady state:

- (a) Faster
- (b) More slowly
- (c) At the same rate
- (d) Oscillatory

Answer: (b)

Explanation: Larger L increases $\tau = L/R$, so current changes more slowly.

Resources: Same as Q11

Q18. Placing a ferromagnetic core inside a coil generally:

- (a) Decreases its inductance
- (b) Increases its inductance
- (c) Leaves inductance unchanged
- (d) Eliminates inductance

Answer: (b)

Explanation: The core increases flux linkage, raising inductance.

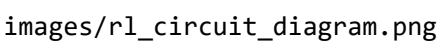
Resources: Same as Q11

Q19. In the RL circuit diagram, which element stores energy in a magnetic field?

- (a) Battery
- (b) Resistor
- (c) Inductor
- (d) Switch

Answer: (c)

Explanation: Energy is stored in the magnetic field of the inductor.

Image: The image is a placeholder for a circuit diagram, likely showing a battery, a resistor, an inductor, and a switch connected in a loop.

Resources: Same as Q11

Q20. Self-inductance is:

- (a) Induction of emf in a coil by its own changing current
- (b) Induction only between two separate coils
- (c) A property unique to resistors
- (d) A purely electrostatic effect

Answer: (a)

Explanation: A changing current changes its own flux and induces emf in the same circuit.

Resources: Same as Q11

AC Circuits I (Q21–Q30)

Q21. A standard AC source voltage is usually written as:

- (a) $V(t) = V_0 t$
- (b) $V(t) = V_0 \sin(\omega t)$
- (c) $V(t) = IR$
- (d) $V(t) = V_0 e^t$

Answer: (b)

Explanation: Ideal sinusoidal sources are written as $V_0 \sin(\omega t)$ (or cosine).

Image: [images/ac_waveform.png](#)

Resources: OpenStax Vol. 2 – 15.2 Simple AC Circuits

Q22. The rms value of $V(t) = V_0 \sin(\omega t)$ is:

- (a) V_0
- (b) $V_0/2$
- (c) $V_0/\sqrt{2}$
- (d) $\sqrt{2}V_0$

Answer: (c)

Explanation: For a sine wave, $V_{\text{rms}} = V_0/\sqrt{2}$.

Resources: Same as Q21

Q23. In a purely resistive AC circuit, current and voltage are:

- (a) In phase
- (b) 90° out of phase
- (c) 180° out of phase
- (d) Unrelated

Answer: (a)

Explanation: They reach maxima and zeros at the same time.

Resources: Same as Q21

Q24. The inductive reactance is:

- (a) $X_L = 1/(\omega C)$
- (b) $X_L = \omega L$
- (c) $X_L = R$
- (d) $X_L = 1/R$

Answer: (b)

Explanation: Inductive opposition to AC grows with frequency: $X_L = \omega L$.

Resources: Same as Q21

Q25. The capacitive reactance is:

- (a) $X_C = 1/(\omega C)$
- (b) $X_C = \omega C$
- (c) $X_C = R$
- (d) $X_C = \omega L$

Answer: (a)

Explanation: X_C decreases as frequency increases: $X_C = 1/(\omega C)$.

Resources: Same as Q21

Q26. In a purely inductive AC circuit, current:

- (a) Leads voltage by 90°
- (b) Lags voltage by 90°
- (c) Is in phase
- (d) Is zero

Answer: (b)

Explanation: Voltage leads current in an inductor.

Resources: Same as Q21

Q27. In a purely capacitive AC circuit, current:

- (a) Leads voltage by 90°
- (b) Lags voltage by 90°
- (c) Is in phase
- (d) Is opposite in phase

Answer: (a)

Explanation: Current leads voltage in a capacitor.

Resources: Same as Q21

Q28. Impedance Z in AC circuits is:

- (a) Only the resistance R
- (b) The complex combination of resistance and reactance
- (c) Always purely imaginary
- (d) Always purely real

Answer: (b)

Explanation: Impedance generalizes resistance to include reactance and phase.

Resources: Same as Q21

Q29. A phasor diagram is primarily used to represent:

- (a) Static fields
- (b) Phase relationships between AC quantities
- (c) Circuit schematics
- (d) Particle trajectories

Answer: (b)

Explanation: Phasors diagrammatically show magnitudes and relative phases.

Resources: Same as Q21

Q30. Rms values are useful because they:

- (a) Equal peak values
- (b) Directly relate to average power like DC values

- (c) Remove all phase information
- (d) Are easier to graph

Answer: (b)

Explanation: They allow using $P = VI$ as in DC circuits.

Resources: Same as Q21

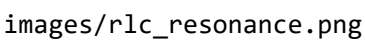
AC Circuits II (Q31–Q40)

Q31. In a series RLC circuit at resonance, the net reactance is:

- (a) Positive
- (b) Negative
- (c) Zero
- (d) Infinite

Answer: (c)

Explanation: At resonance $X_L = X_C$, so the reactive parts cancel.

Image: 

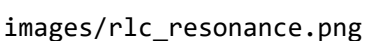
Resources: OpenStax Vol. 2 – 15.2 context

Q32. At resonance in a series RLC circuit, the current for a given source is:

- (a) Minimum
- (b) Maximum
- (c) Zero
- (d) Undefined

Answer: (b)

Explanation: Impedance is minimal ($Z = R$), so current amplitude is largest.

Image: 

Q33. The resonant angular frequency ω_0 of a series RLC circuit is:

- (a) $1/RC$
- (b) $1/\sqrt{LC}$
- (c) R/L
- (d) $\sqrt{LC}$

Answer: (b)

Explanation: At $\omega_0 = 1/\sqrt{LC}$, $X_L = X_C$.

Image: images/rlc_resonance.png

Q34. The quality factor Q of a resonance describes:

- (a) Only resistance
- (b) Sharpness (selectivity) of the resonance peak
- (c) Circuit size
- (d) Battery voltage

Answer: (b)

Explanation: Higher Q means a narrower, more selective resonance.

Image: images/rlc_resonance.png

Q35. A power factor near 1 indicates:

- (a) Mostly reactive power
- (b) Mostly real (useful) power
- (c) No power delivered
- (d) Infinite impedance

Answer: (b)

Explanation: Voltage and current are nearly in phase, so power transfer is efficient.

Q36. Capacitors are added to inductive loads mainly to:

- (a) Increase frequency
- (b) Correct (improve) power factor

- (c) Increase resistance
- (d) Eliminate real power

Answer: (b)

Explanation: Capacitors supply reactive power that cancels part of the inductive reactance.

Q37. In an ideal transformer, V_s/V_p equals:

- (a) N_p/N_s
- (b) N_s/N_p
- (c) I_p/I_s
- (d) I_s/I_p

Answer: (b)

Explanation: For an ideal transformer, $V_s/V_p = N_s/N_p$.

Q38. For an ideal transformer (neglecting losses):

- (a) Input power equals output power
- (b) Input power is twice output power
- (c) Output power is zero
- (d) Power is not conserved

Answer: (a)

Explanation: Conservation of energy requires $P_p \approx P_s$.

Q39. Above resonance, a series RLC circuit behaves mainly:

- (a) Capacitive
- (b) Inductive
- (c) Resistive
- (d) Superconducting

Answer: (b)

Explanation: For $\omega > \omega_0$, $X_L > X_C$, giving net inductive reactance.

Q40. Below resonance, a series RLC circuit behaves mainly:

- (a) Capacitive
- (b) Inductive
- (c) Resistive
- (d) Superconducting

Answer: (a)

Explanation: For $\omega < \omega_0$, $X_C > X_L$, giving net capacitive reactance.

Maxwell's Equations & EM Waves (Q41–Q50)

Q41. Gauss's law for magnetism states:

- (a) $\nabla \cdot \vec{B} = \mu_0 \rho_m$
- (b) Net magnetic flux through any closed surface is zero
- (c) Magnetic field lines start on magnetic charges
- (d) Magnetic field is always uniform

Answer: (b)

Explanation: There are no isolated magnetic monopoles; flux in equals flux out.

Resources: OpenStax Vol. 2 – 16.1

Q42. In vacuum, the speed of light is:

- (a) $c = \mu_0 \epsilon_0$
- (b) $c = 1/(\mu_0 \epsilon_0)$
- (c) $c = 1/\sqrt{\mu_0 \epsilon_0}$
- (d) $c = \sqrt{\mu_0/\epsilon_0}$

Answer: (c)

Explanation: Maxwell showed EM waves propagate at $c = 1/\sqrt{\mu_0 \epsilon_0}$.

Q43. In an electromagnetic plane wave:

- (a) $\vec{E} \parallel \vec{B} \parallel$ propagation
- (b) $\vec{E} \perp \vec{B}$ and both $\perp$ propagation
- (c) $\vec{E}$ only along propagation
- (d) $\vec{B}$ only along propagation

Answer: (b)

Explanation: EM waves are transverse: E , B , and direction are mutually perpendicular.

Q44. The displacement current term was added to Ampere's law to:

- (a) Change Gauss's law
- (b) Match Faraday's law
- (c) Ensure consistency with charge conservation in time-varying fields
- (d) Remove magnetic fields

Answer: (c)

Explanation: It allows Ampere–Maxwell law to hold where electric fields change but no conduction current flows.

Q45. The Poynting vector $\vec{S}$ represents:

- (a) Charge density
- (b) Energy flux (power per unit area) of EM waves
- (c) Magnetic monopole current
- (d) Phase difference

Answer: (b)

Explanation: $\vec{S}$ points in the direction of energy transport.

Q46. Electromagnetic waves carry:

- (a) Only electric energy
- (b) Only magnetic energy

- (c) Both energy and momentum
- (d) Neither

Answer: (c)

Explanation: They can transfer energy and exert radiation pressure.

Q47. For a plane EM wave in free space, E_0/B_0 equals:

- (a) 1
- (b) c
- (c) $1/c$
- (d) 0

Answer: (b)

Explanation: Fields are related by $E_0 = cB_0$.

Q48. A linearly polarized EM wave is one in which:

- (a) Frequency changes with time
- (b) The electric field oscillates along a fixed direction
- (c) Direction of propagation changes randomly
- (d) The magnetic field is zero

Answer: (b)

Explanation: Polarization refers to a fixed direction of the electric field vector.

Q49. Most radio-frequency EM waves are generated by:

- (a) Static charges
- (b) Accelerating charges or oscillating currents
- (c) Permanent magnets
- (d) Nuclear reactions only

Answer: (b)

Explanation: Time-varying currents radiate EM waves.

Q50. The diagram showing perpendicular $\vec{E}$ and $\vec{B}$ traveling together represents:

- (a) Sound wave
- (b) Water wave
- (c) Electromagnetic plane wave
- (d) Standing wave on a string

Answer: (c)

Explanation: Mutually perpendicular E and B fields propagating together define an EM wave.

Image: [images/em_wave.png](#)

Reflection & Refraction (Q51–Q60)

Q51. Snell's law is:

- (a) $n_1 \sin \theta_1 = n_2 \sin \theta_2$
- (b) $F = ma$
- (c) $V = IR$
- (d) $pV = nRT$

Answer: (a)

Explanation: It relates angles of incidence and refraction via indices of refraction.

Resources: OpenStax Vol. 3 – 1.3 Refraction

Q52. When light enters a medium with higher n , it:

- (a) Speeds up and bends away from the normal
- (b) Slows down and bends toward the normal
- (c) Slows down and bends away from the normal
- (d) Is unchanged

Answer: (b)

Explanation: Higher index means lower speed and bending toward the normal.

Q53. Total internal reflection occurs when light:

- (a) Goes from low n to high n
- (b) Goes from high n to low n with $\theta > \theta_c$
- (c) Strikes any mirror
- (d) Travels in vacuum

Answer: (b)

Explanation: Above the critical angle, all light reflects back into the denser medium.

Q54. For a plane mirror, the angle of reflection equals:

- (a) Angle of refraction
- (b) Angle of incidence
- (c) Critical angle
- (d) 90°

Answer: (b)

Explanation: Law of reflection: $\theta_r = \theta_i$.

Q55. The index of refraction is:

- (a) $n = v/c$
- (b) $n = c/v$
- (c) $n = 1/v$
- (d) $n = c$

Answer: (b)

Explanation: n compares light speed in vacuum to that in the medium.

Q56. If $n_2 < n_1$, the refracted ray:

- (a) Bends toward the normal
- (b) Bends away from the normal
- (c) Always totally internally reflects
- (d) Disappears

Answer: (b)

Explanation: Lower index gives a larger angle from the normal.

Q57. At Brewster's angle, the reflected light is:

- (a) Unpolarized
- (b) Circularly polarized
- (c) Linearly polarized
- (d) Completely absorbed

Answer: (c)

Explanation: The reflected ray becomes fully polarized in one plane.

Q58. A ray entering along the normal ($\theta = 0$) into a new medium:

- (a) Bends toward the surface
- (b) Bends away from the surface
- (c) Continues straight
- (d) Totally reflects

Answer: (c)

Explanation: With $\theta_1 = 0$, Snell's law gives $\theta_2 = 0$, so no bending.

Q59. A correct refraction diagram from air to glass shows the refracted ray:

- (a) Bending toward the normal
- (b) Bending away from the normal
- (c) Parallel to the surface
- (d) Returning to air

Answer: (a)

Explanation: Since $n_{\text{glass}} > n_{\text{air}}$, the ray bends toward the normal.

Image: images/refraction_diagram.png

Q60. Angles of incidence and refraction are always measured:

- (a) From the surface
- (b) From the normal
- (c) From the vertical
- (d) Arbitrarily

Answer: (b)

Explanation: Measuring from the normal is required for Snell's law.